

LILA F. D. PERKINS

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July 8, 2026

EDUCATION

University of Washington

Ph.D. Student in Applied Mathematics

Advisor: Baosen Zhang (Department of Electrical & Computer Engineering)

Sep 2025 – Present

Seattle, WA

University of Washington

Master of Science in Applied and Computational Mathematics

GPA: 3.90/4.00

Jun 2026

Seattle, WA

University of Michigan

Bachelor of Science in Honors Mathematics with High Distinction

Minor in Complex Systems

College of Literature, Science, and the Arts (LSA) Residential College and Honors Program

GPA: 3.98/4.00

May 2025

Ann Arbor, MI

Budapest Semesters in Mathematics (BSM)

Completed with Honors Recognition

Competitive semester program emphasizing the Hungarian method of mathematics instruction.

Coursework: Advanced Combinatorics, Theory of Computing, Non-Euclidean Geometries.

GPA: 4.0/4.0

Aug – Dec 2023

Budapest, Hungary

RESEARCH EXPERIENCE

University of Washington

Graduate Student Researcher, Dept. of Applied Mathematics

Advisor: Prof. Baosen Zhang (Dept. of Electrical and Computer Engineering)

Sep 2025 – Present

Seattle, WA

- Budget-Constrained Resource Allocation: Proved the existence and uniqueness of competitive equilibrium in electricity markets with budget constrained users; demonstrated equivalence to a convex welfare maximization problem with modified utility functions (work submitted to *IEEE Control Systems Letters* and *IEEE CDC 2026*).
- Budget Constraints in Networked Markets: Currently extending frameworks to model strategic data centers participating in both compute and electricity markets across multiple geographic regions; analyzing conditions for equilibrium existence and efficiency loss in Cournot settings.

Park City Mathematics Institute/Institute for Advanced Study

Undergraduate Summer School

- Conducted research exploring eigenpolytopes as a bridge between abstract algebra and graph theory within a three-week program on Extremal and Probabilistic Combinatorics.

July 2025

Park City, UT

National Renewable Energy Laboratory

DOE Science Undergraduate Laboratory Internship

Advisor: Dr. Bernard Knueven

Jun – Aug 2024

Golden, CO

- Grid Resilience: Developed a computationally efficient numerical linear algebra approach using the Woodbury Matrix Identity to identify potential new transmission lines that improve grid resilience with respect to contingency analyses.
- Implementation: Implemented solutions in Julia and tested on the IEEE Power Grid Library; presented at the NREL Intern Symposium and the University of Michigan SIAM Student Mini-Symposium.

University of Michigan

MATH 389 Explorations in Mathematics

Advisors: Prof. Aaron Pixton and Dr. Patrick Boland

Winter 2024

Ann Arbor, MI

- Analyzed strategies for the Taxman number factoring game using number theory, graph theory, and computational methods.

- Explored patterns arising from agent motion on varying graph structures using algebraic and computational methods.

National Renewable Energy Laboratory

DOE Science Undergraduate Laboratory Internship

Advisor: Dr. Gayathri Krishnamoorthy

Jun – Aug 2023

Golden, CO

- Resilience Planing: Developed a grid resilience database and a FEMA Expected Annual Loss calculator for utility service territories to inform infrastructure investment strategies.
- Dissemination: Co-authored three DOE Technical Reports on distribution utility resilience planning for wild-fires, hurricanes, and winter storms.

PUBLICATIONS

Perkins, L. and Zhang, B. “Resource Allocation in Electricity Markets with Budget Constrained Customers.” *IEEE Control Systems Letters*, June 2026. [DOI: 10.1109/LCSYS.2026.3704356](https://doi.org/10.1109/LCSYS.2026.3704356)

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. “Current Practices in Distribution Utility Resilience Planning for Wildfires.” DOE Technical Report, August 2024. [DOI:10.2172/2478838](https://doi.org/10.2172/2478838)

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. “Current Practices in Distribution Utility Resilience Planning for Hurricanes and Non-Winter Storms.” DOE Technical Report, August 2024. [DOI:10.2172/2478839](https://doi.org/10.2172/2478839)

Keen, J., Matsuda-Dunn, R., Krishnamoorthy, G., Clapper, H., **Perkins, L.**, Leddy, L., and Grue, N. “Current Practices in Distribution Utility Resilience Planning for Winter Storms.” DOE Technical Report, August 2024. [DOI:10.2172/2478837](https://doi.org/10.2172/2478837)

PRESENTATIONS

Perkins, L. and Knueven, B. “A mathematical approach for identifying new transmission lines to enhance grid resilience.” University of Michigan SIAM Student Mini-Symposium, Ann Arbor, MI. September 2024. [Poster]

Perkins, L. and Knueven, B. “A mathematical approach for identifying new transmission lines to enhance grid resilience.” NREL Intern Symposium, Golden, CO. August 2024. [Poster]

Perkins, L. and Krishnamoorthy, G. “Grid resilience investment database: supplement to Distribution resilience framework for hazard cases.” NREL Intern Symposium, Golden, CO. August 2023. [Poster]

AWARDS AND HONORS

Clean Energy Institute Graduate Fellowship 2026 - 2027
Awarded for proposed research on optimization methods for scheduling compute workloads across heterogeneous infrastructure and data centers’ potential to price couple regional electricity markets.

DOE Computational Science Graduate Fellowship 2026 - 2030
Awarded for proposed research on optimization methods for scheduling compute workloads across heterogeneous, geographically distributed infrastructure.

NSF Graduate Research Fellowship Apr 2026
Awarded for proposed research on optimization methods for AI workloads and electricity market integration.

Boeing Graduate Fellowship 2025 – 2027
University of Washington, Department of Applied Mathematics.

Phi Beta Kappa Winter 2025
University of Michigan.

Mathematics Merit Scholarship 2024 – 2025
University of Michigan, Department of Mathematics.

Phi Beta Phi Winter 2023
University of Michigan.

James B. Angell Scholar	2023 – 2025
University of Michigan. Awarded to students earning all A or A+ grades in a full-credit semester.	
J. Perry Bartlett STEM Scholar	2021 – 2025
Four-year college scholarship to school of choice.	

TEACHING AND EMPLOYMENT

University of Michigan, Department of Mathematics	Winter 2025
<i>Course Assistant — MATH 389: Explorations in Mathematics</i>	<i>Ann Arbor, MI</i>
Mentored groups of undergraduates on mathematics research projects.	
University of Michigan, Department of Mathematics	Winter 2025
<i>Grader — MATH 404: Intermediate Differential Equations</i>	<i>Ann Arbor, MI</i>
Graded weekly homework for undergraduate differential equations course.	
University of Michigan, Department of Mathematics	2022 – 2023
<i>Peer Tutor — MATH 217: Linear Algebra</i>	<i>Ann Arbor, MI</i>
Provided one-on-one tutoring for undergraduate linear algebra students.	
BrightLine Group	Jul – Oct 2022
<i>Data Analysis Intern</i>	<i>Boulder, CO</i>
Analyzed residential solar installation efficiency and evaluated California's Single-Family Affordable Solar Homes (SASH) and Disadvantaged Communities – Single-Family Solar Homes (DAC-SASH) programs effectiveness. Drafted methodology section of client report.	

SERVICE AND OUTREACH

Outdoors for All	Winter 2025
<i>Volunteer Adaptive Ski Instructor</i>	<i>Seattle, WA</i>
Teaching adaptive skiing to participants with physical and/or cognitive disabilities through a Seattle-based nonprofit making outdoor recreation accessible.	
Special Olympics Michigan, Area 20	2022 – 2025
<i>Volunteer</i>	<i>Ann Arbor, MI</i>
Assisted with weekly basketball practices for athletes with intellectual disabilities.	
Ignite Adaptive Sports	2019 – 2021
<i>Volunteer Junior Instructor</i>	<i>Boulder, CO</i>
Taught adaptive skiing to participants with physical and/or cognitive disabilities.	

SELECTED ADVANCED COURSEWORK

University of Washington
Optimization: Fundamentals and Applications; Networks and Combinatorial Optimization; Numerical Linear Algebra; Numerical Analysis; Numerical Methods for Partial Differential Equations
University of Michigan (all graduate-level)
Computer Modeling of Complex Systems; Theory of Complex Systems; Numerical Methods for Differential Equations; Probability Theory; Sustainable Energy Systems; Advanced Energy Solutions

TECHNICAL SKILLS AND LANGUAGES

Programming: Julia, Python, MATLAB, C++, Mathematica, LaTeX, Excel, Igor Pro, SOLIDWORKS, AutoCAD, NLR System Advisor Model
Languages: English (native); Spanish (fluent); Portuguese (proficient); Hungarian (basic)

PROFESSIONAL MEMBERSHIPS

American Mathematical Society (AMS)	2026 – Present
Institute of Electrical and Electronics Engineers (IEEE), Graduate Student Member	2026 – Present
IEEE Power & Energy Society (PES)	
Society for Industrial and Applied Mathematics (SIAM)	2026 – Present